



Electrical Machines

Detailed Solutions • 2013–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

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PrepFusionTM

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Electrical Machines (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7	3	Unit (chapter) number — Unit III of this book
	24	GATE exam year — 2024 (two digits; 98 = 1998)
	7	Question number within that unit and year

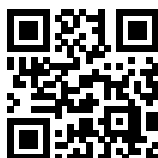
So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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SOLUTIONS



Magnetic Circuits

Topics

1. Magnetically Coupled Circuits, Dot Convention
2. Energy in Coupled circuits
3. Electric and Magnetic Materials

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Q1.25.1

MCQ

2025

1M

Ans: A

☒ ☒ ☐

Given circuit is in a magnetically opposing dot configuration. By modeling the magnetically coupled coils into its equivalent T-network:

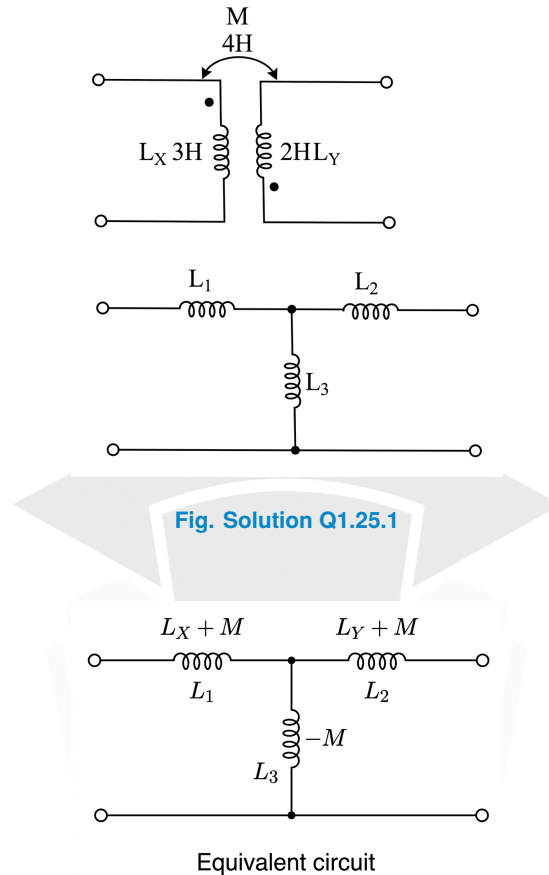


Fig. Solution Q1.25.1

For a mutually coupled inductor pairs with opposing dots, the mutual terms add up positively in the series arms of the equivalent T-network ($L_1 = L_X + M$, $L_2 = L_Y + M$) and negatively in the common shunt branch ($L_3 = -M$). The values of the equivalent T-model inductances can be derived directly by analyzing the network:

$$L_1 = L_X + M = 3 + 4 = 7 \text{ H}$$

$$L_2 = L_Y + M = 2 + 4 = 6 \text{ H}$$

$$A) L_1 = 7 \text{ H}, \quad L_2 = 6 \text{ H}, \quad L_3 = -4 \text{ H}$$

Q1.24.1

NAT

2024

2M

Ans: 0.25

☒ ☒ ☐

Let the two self-inductances be equal such that:

$$L_1 = L_2 = L$$

When connected in a series-aiding configuration, the equivalent total inductance is given by:

$$L_1 + L_2 + 2M = 500 \text{ mH} \implies 2L + 2M = 500 \text{ mH} \quad \text{--- (1)}$$

When connected in a series-opposing configuration, the equivalent total inductance is given by:

$$L_1 + L_2 - 2M = 300 \text{ mH} \implies 2L - 2M = 300 \text{ mH} \quad \text{--- (2)}$$

Adding equations (1) and (2):

$$2(L_1 + L_2) = 800 \text{ mH} \implies 2(2L) = 800 \text{ mH}$$

$$L = L_1 = L_2 = 200 \text{ mH}$$

Substituting $L = 200 \text{ mH}$ back into equation (1):

$$2(200) + 2M = 500 \implies 2M = 100 \implies M = 50 \text{ mH}$$

The coupling coefficient K is calculated as:

$$K = \frac{M}{\sqrt{L_1 L_2}} = \frac{M}{L} = \frac{50 \text{ mH}}{200 \text{ mH}} = \frac{1}{4} = 0.25$$

$$\boxed{0.25}$$

Q1.23.1

NAT

2023

2M

Ans: 17.37



Given,

$$K = 0.6$$

$$L_1 = 4 \text{ H}, \quad L_2 = 5 \text{ H}$$

$$i_1(t) = 1 \sin(200\pi t) \text{ mA} \implies I_{1P} = 1 \text{ mA}$$

$$i_2(t) = 2 \sin(200\pi t) \text{ mA} \implies I_{2P} = 2 \text{ mA}$$

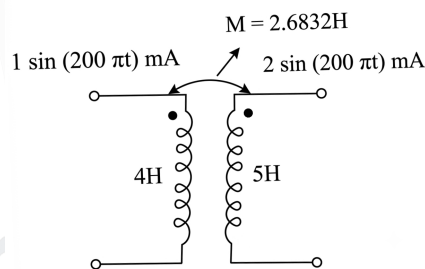


Fig. Solution Q1.23.1

Mutual inductance M :

$$M = K\sqrt{L_1 L_2} = 0.6\sqrt{4 \times 5} = 0.6\sqrt{20} \approx 2.6832 \text{ H}$$

The maximum peak energy stored in the mutually coupled circuit occurs when both currents reach their peak values simultaneously (since they are in phase). Looking at the dot polarity, both currents enter the dots, meaning the magnetic fields aid each other:

$$E_{\text{Peak}} = \frac{1}{2}L_1 I_{1P}^2 + \frac{1}{2}L_2 I_{2P}^2 + M I_{1P} I_{2P}$$

Substituting peak values in milliamperes ($1 \text{ mA} = 10^{-3} \text{ A}$):

$$E_{\text{Peak}} = \frac{1}{2}(4)(1 \times 10^{-3})^2 + \frac{1}{2}(5)(2 \times 10^{-3})^2 + 2.6832(1 \times 10^{-3})(2 \times 10^{-3})$$

$$E_{\text{Peak}} = [2 \times 10^{-6}] + [10 \times 10^{-6}] + [5.3664 \times 10^{-6}] \text{ J}$$

$$E_{\text{Peak}} = (2 + 10 + 5.3664) \mu\text{J} = 17.3664 \mu\text{J}$$

Rounding off to two decimal places:

$$\boxed{17.37}$$

Key Points

- Total instantaneous energy stored in coupled coils: $E(t) = \frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 \pm Mi_1i_2$
- The sign of the M term is positive if both currents enter (or both exit) the dot terminals.

Mistakes to be avoided

- Do not forget to convert milliamperes to amperes ($\times 10^{-3}$), which becomes 10^{-6} (micro) when squared or multiplied.

Q1.15.1

MCQ

2015

1M

Ans: B

● ○ ○

From the ideal transformer voltage ratio relations:

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$\frac{V_2}{V_1} = \frac{1}{N}$$

$$V_2 = NV_1 \Rightarrow NV_i \sin \omega t$$

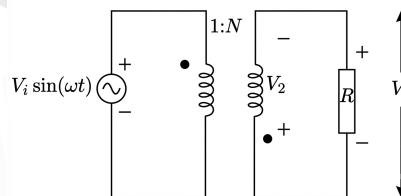


Fig. Solution Q1.15.1

According to the given dot polarity convention, the current enters the dot on the primary winding while the induced voltage is positive at the terminal opposite to the dot on the secondary winding. Thus, the polarity of V_2 is inverted relative to the marked terminal orientation of V_o :

$$V_o = -NV_i \sin \omega t$$

$$B) - NV_i \sin \omega t$$

Q1.13.1

MCQ

2013

2M

Ans: B

● ● ●

We evaluate the circuit arrangement under two distinct excitation setups:

Case 1: $V_{WX1} = 100 \text{ V}$ is applied across WX terminals. The secondary side voltage of the ideal transformer with a turns ratio of 1 : 1.25 becomes:

$$V_{\text{secondary}} = 100 \times 1.25 = 125 \text{ V}$$

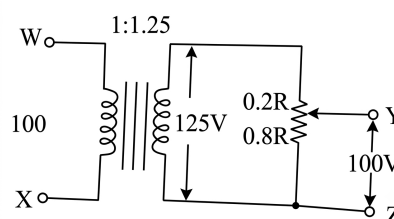


Fig. Solution Q1.13.1

The attenuator network splits the resistance into a $0.2R$ and $0.8R$ potentiometer configuration. The open-circuit voltage across the output terminals YZ is taken across the $0.8R$ section:

$$V_{yz1} = 125 \times \frac{0.8R}{0.2R + 0.8R} = 125 \times 0.8 = 100 \text{ V}$$

Hence, the first ratio is:

$$\frac{V_{yz1}}{V_{wx1}} = \frac{100}{100}$$

Case 2: $V_{yz2} = 100 \text{ V}$ is applied backward across YZ terminals. Since it is an ideal transformer, it draws zero magnetizing current to set up the rated flux ($i = 0$). Consequently, there is no current flow and no voltage drop across the upper attenuator resistor segment $0.2R$:

$$V_{\text{secondary}} = V_{yz2} = 100 \text{ V}$$

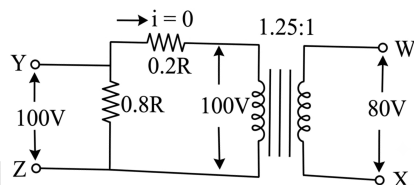


Fig. Solution Q1.13.1

Stepping down the voltage to the primary winding using the effective 1.25 : 1 reverse turns relation:

$$V_{wx2} = \frac{100}{1.25} = 80 \text{ V}$$

Hence, the second ratio is:

$$\frac{V_{wx2}}{V_{yz2}} = \frac{80}{100}$$

B) 100/100 and 80/100

Q1.13.1

MCQ

2013

2M

Ans: C

● ○ ○

The individual quality factors (Q -factors) for the two uncoupled inductive coils at a chosen operating angular frequency ω for a series RL circuit are defined as:

$$q_1 = \frac{\omega L_1}{R_1}, \quad q_2 = \frac{\omega L_2}{R_2}$$

When these two coils are connected in series, the total equivalent inductance L_t and total equivalent resistance R_t become:

$$L_t = L_1 + L_2$$

$$R_t = R_1 + R_2$$

The effective quality factor q_t of the series combination at the same operating frequency is given by:

$$q_t = \frac{\omega(L_1 + L_2)}{R_1 + R_2}$$

Substituting $\omega L_1 = q_1 R_1$ and $\omega L_2 = q_2 R_2$ into the equation:

$$q_t = \frac{q_1 R_1 + q_2 R_2}{R_1 + R_2}$$

C) $(q_1 R_1 + q_2 R_2) / (R_1 + R_2)$

DEBUG INFO

Chapter I

Magnetic Circuits

Total Questions : 6

MCQ : 4

MSQ : 0

NAT : 2

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SOLUTIONS



Single Phase Transformers

Topics

1. Equivalent Circuit and Phasor Diagram
2. Open Circuit and Short Circuit Tests
3. Voltage Regulation and Efficiency of Transformer

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Q2.26.1 MCQ 2026 1M Ans: C ☒ ☐ ☐

The secondary impedance referred to the primary side is:

$$Z'_2 = \frac{Z_2}{K^2}, \quad \left(K = \frac{N_2}{N_1}\right)$$

$$Z'_2 = \left(\frac{N_1}{N_2}\right)^2 \cdot Z_2$$

For a step-down transformer:

$$\frac{N_1}{N_2} > 1$$

$$\therefore Z'_2 > Z_2$$

$$\boxed{C) Z'_2 > Z_2}$$

Q2.26.2 NAT 2026 2M Ans: 2490 to 2510 ☒ ☒ ☐

Given parameters:

$$\cos \phi = \frac{P}{VI} = \frac{363}{220 \times 2.75} = 0.6$$

The magnetizing current component is:

$$I_m = I_0 \sin \phi = 2.75 \times 0.8 = 2.2 \text{ A}$$

The magnetizing reactance X_m relative to the low-voltage side is:

$$X_m = \frac{V}{I_m} = \frac{220}{2.2} = 100 \Omega$$

Referred to the high-voltage side, the magnetizing reactance becomes:

$$X'_m = \frac{X_m}{K^2} = \frac{100}{\left(\frac{220}{1100}\right)^2}$$

$$= \left(\frac{1100}{220}\right)^2 \times 100 = 2500 \Omega$$

$$\boxed{2500}$$

Q2.25.1 NAT 2025 2M Ans: 31.00 to 31.50 ☒ ☒ ☐

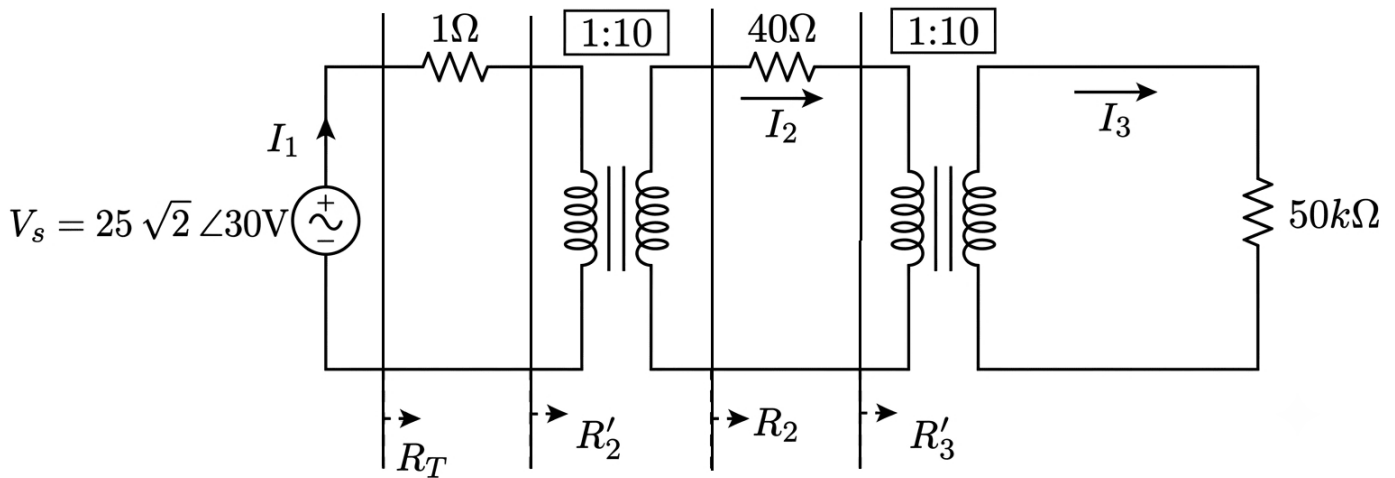


Fig. Solution Q2.25.1

The transformer network can be systematically referred across stages by squaring the respective turns ratios. Referring the loads back step-by-step simplifies the multi-stage network into a single equivalent loop:

$$R_{eq} = 1\Omega + 4\Omega = 5\Omega$$

The loop current is found using the RMS voltage of the source:

$$I = \frac{25}{10} (\text{rms}) \Rightarrow I = 2.5\text{ A}$$

The power dissipated in the load is:

$$P_{50k\Omega} = P_{5\Omega} = 2.5^2 \times 5 = 31.25\text{ W}$$

31.25

Q2.24.1

NAT

2024

2M

Ans: 95.2 to 95.4

☒ ☒ ☐

Given parameters: Rating = 50 kVA, $\eta_{FL, UPF} = 95\%$, pf = 1 and Core losses = neglected ($W_i = 0$)

Using the full-load efficiency equation at unity power factor:

$$W_{cu}(x) = x^2 W_{cu, FL}$$

$$\text{Efficiency} = \frac{\text{Output Power}}{\text{Output Power} + \text{Iron Loss} + \text{Copper Loss}}$$

$$\% \eta_{FL, UPF} = \frac{50 \times 10^3 \times 1 \times 1}{(50 \times 10^3 \times 1 \times 1) + (W_i = 0) + W_{cu, FL}}$$

$$0.95 = \frac{50 \times 10^3}{(50 \times 10^3) + W_{cu, FL}} \Rightarrow W_{cu, FL} = 2631.57\text{ W}$$

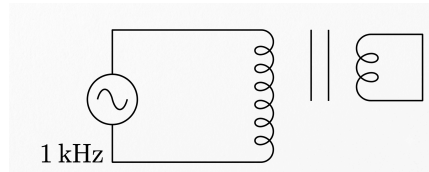
Now computing efficiency at 75% full load and 0.8 power factor lagging:

$$\begin{aligned} \% \eta_{75\%FL, 0.8pf} &= \frac{(0.75 \times 50 \times 10^3) \times 0.8}{(0.75 \times 50 \times 10^3 \times 0.8) + 0 + (0.75)^2 \times 2631.57} \times 100 \\ &= 95.29\% \end{aligned}$$

95.2 to 95.4

Q2.23.1
NAT
2023
2M
Ans: 1
• • •

Given short-circuit test data at $f = 1 \text{ kHz}$: $w_{sc} = 8 \text{ W}$, $V_{sc} = 6 \text{ V}$, $I_{sc} = 2 \text{ A}$


Fig. Solution Q2.23.1

The short-circuit impedance and equivalent parameters referred to the primary are:

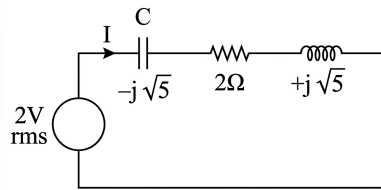
$$Z_{sc} = \frac{V_{sc}}{I_{sc}} = \frac{6}{2} = 3 \Omega$$

$$w_{sc} = I_{sc}^2 R_{01} \Rightarrow R_{01} = \frac{8}{2^2} = 2 \Omega$$

$$Z_{sc}^2 = R_{01}^2 + X_{01}^2 \Rightarrow X_{01}^2 = 3^2 - 2^2 = 5$$

$$X_{01} = \sqrt{5} \Rightarrow X_{01} = +j\sqrt{5}$$

When connected with a capacitor C across a 2 V source:


Fig. Solution Q2.23.1

The capacitance value is:

$$C = \frac{1}{2\pi\sqrt{5}} \text{ mF}$$

The capacitive reactance is:

$$-jX_c = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 10^3 \times \frac{1}{2\pi\sqrt{5}} \times 10^{-3}} = -jX_c = \sqrt{5}$$

Evaluating the final circuit loop at resonance: Since $X_L = X_C$, the circuit is purely resistive and under a resonance condition:

$$I = \frac{V}{R} = \frac{2}{2} = 1 \text{ A}$$

1

Q2.21.1
NAT
2021
1M
Ans: 0.24 to 0.26
☒ ☐ ☐

Given parameters: $L_0 = 250 \times 10^{-3} \text{ H}$, $R_0 = 300 \Omega$ The inductive reactance is:

$$X_{L_0} = 2\pi f L_0 = 2\pi \times 50 \times 250 \times 10^{-3} = 78.54 \Omega$$

The susceptance and conductance are:

$$B_{L_0} = \frac{1}{X_{L_0}} = \frac{1}{78.54} = 0.01273 \text{ U}$$

$$G_0 = \frac{1}{300} = 3.33 \times 10^{-3} \text{ U}$$

The total admittance in parallel circuit is:

$$Y_0 = \sqrt{G_0^2 + B_{L_0}^2} = 0.01316 \text{ U}$$

The no-load power factor is:

$$\cos \phi_0 = \frac{G_0}{Y_0} = 0.253 \quad (\text{lag})$$

Q2.21.2
MCQ
2021
2M
Ans: A
☒ ☐ ☐

Given parameters: $\eta_{\max} = 98\%$, $W_{\text{iron}} = 80 \text{ W}$, $R_{01} = 0.5 \Omega$, $I_1 = 25 \text{ A}$ At maximum efficiency, the variable copper loss equals the constant iron loss (x is the loading fraction):

$$x^2 I_1^2 R_{01} = W_i$$

$$x^2 (25)^2 (0.5) = 80$$

$$x = 50.59\%$$

Thus, maximum efficiency occurs at 50.59% of full load.

DEBUG INFO

Chapter II

Single Phase Transformers

Total Questions : 7

MCQ : 2

MSQ : 0

NAT : 5

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SOLUTIONS



Three Phase Induction Machines

Topics

1. Principle of Operation , Types and Performance
2. Equivalent Circuit of Induction Motor
3. Torque-Slip, Torque-Speed Characteristics
4. Stability and Testing – No-Load (OC) and Blocked-Rotor (SC) Tests
5. Starting of Induction Motors
6. Speed Control
7. Types of losses and efficiency calculation of Electric Machines

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Q3.25.1

MCQ

2025

2M

Ans: A

☒ ☒ ☐

Given conditions:

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{2} = 3000 \text{ rpm}$$

The initial slip is:

$$s_1 = \frac{3000 - 2970}{3000} = 0.01$$

In the normal operating (low-slip) region of a standard induction motor, torque varies linearly with slip.

$$T_1 \propto s_1$$

As torque is doubled, $T_2 = 2T_1$, meaning the slip gets doubled as well:

$$s_2 = 2s_1 = 2 \times 0.01 = 0.02$$

Evaluating the new rotor speed N_2 :

$$s_2 = \frac{N_s - N_2}{N_s} \implies \frac{3000 - N_2}{3000} = 0.02$$

$$3000 - N_2 = 60 \implies N_2 = 2940 \text{ rpm}$$

A) 2940

Q3.24.1

NAT

2024

2M

Ans: 17.00 to 17.30

☒ ☒ ☐

Given:

$$T_{st} = 100\% \text{ of } T_{FL} \implies \frac{T_{st}}{T_{FL}} = 1$$

$$T_{max} = 300\% \text{ of } T_{FL} \implies \frac{T_{max}}{T_{FL}} = 3$$

$$s_{T_{max}} = ?$$

The ratio of starting torque to maximum torque is a specific function of the parameter a , where a corresponds to the slip at which maximum torque occurs (s_m).

Using the torque ratios:

$$\therefore \frac{T_{st}/T_{FL}}{T_{max}/T_{FL}} = \frac{1}{3} \implies \frac{T_{st}}{T_{max}} = \frac{1}{3} = \frac{2a}{1+a^2}$$

$$\frac{2a}{1+a^2} = \frac{1}{3} \implies a^2 + 1 - 6a = 0$$

$$a^2 - 6a + 1 = 0$$

Solving for a (slip at maximum torque):

$$a = 0.1715 = 17.15\%$$

17.15

Q3.23.1 NAT 2023 1M Ans: 0 ☒ ☐ ☐

Output voltage when machine is rotated in the clockwise direction is 200 V. If it is rotated in the opposite direction, the armature current direction and shunt field current direction will be changed. Then ϕ_{sh} opposes the $\phi_{residual}$. Net EMF will become zero.

0

Q3.22.1 NAT 2022 1M Ans: 1440 ☒ ☐ ☐

Given parameters: $V = 440 \text{ V}$, $P = 8 \text{ kW}$, Poles = 4, $f = 50 \text{ Hz}$, $S = 0.04$ The synchronous speed is calculated as:

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

The actual rotor speed N_r is:

$$N_r = N_s(1 - s) = 1500(1 - 0.04) = 1440 \text{ rpm}$$

1440

Q3.21.1 MCQ 2021 2M Ans: A ☒ ☒ ☐

By adding external resistance in the rotor circuit of a slip ring induction motor, the total rotor resistance increases, which subsequently shifts the maximum torque point and increases the starting torque. This improve the starting performance of a wound-rotor induction motor by bringing the starting torque closer to the maximum available breakdown torque.

A) The starting torque would be higher.

DEBUG INFO

Chapter III

Three Phase Induction Machines

Total Questions : 5

MCQ : 2

MSQ : 0

NAT : 3

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